



Final exam

Mathematics 1 (Technische Universiteit Eindhoven)



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Exercises

1	2	3	4	5	6
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Surname, First name

5EZB0 Math 2
Final exam Q4

1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9
0	0	0	0	0	0	0

a	☒	c	d	e	f	→ b
a	b	☒	d	e	f	→ c
☒	b	c	☒	e	f	→ a

Answer multiple-choice questions as shown in the example.

Particular Ans on paper exam instructions

- Write in a black or blue pen.
- You answer open-ended questions by using the text box. Provide your answers on the papers inside the answer box underneath a question. **If you need more space for your answers, use the extra space at the end of the exam, and clearly indicate there which question you continue answering. In the text box of the particular question, clearly state that you proceed with your answer on a different page.**
- **The solutions should be formulated clearly, arranged orderly, and should include clear explanations of all steps that are needed to obtain the final answer.**
- Hand in all pages. Do not remove the staple. If you remove it anyhow, check that you hand in all pages.

Dear student,

This is the final exam for 5EZB0. Write down your name and your student ID at the appropriate places above. Make sure that you enter your student ID by fully coloring the appropriate boxes. On the examination attendance card, you fill in the PDF number. You can find the correct number on the top of the first page of your exam (e.g. 1234.pdf).

Please read the following information carefully:

Date exam: 24/06/2024

Start time: 9:00

End time: 12:00 (+30 minutes for time extension students)

Number of questions: 5

Maximum number of points/distribution of points over questions: $80=17+20+15+18+10$

Contribution to the final grade = (number of points)%

Permitted examination aids

- Scrap paper (fully blank)
- Calculator(**non-graphical/ non-programmable**)

Important:

- You are only permitted to visit the toilets under supervision
- Examination scripts (fully completed examination paper, stating name, student number, etc.) must always be handed in
- The house rules must be observed during the examination
- The instructions of subject experts and invigilators must be followed
- Keep your work place as clean as possible: put pencil case and breadbox away, limit snacks and drinks
- You are not permitted to share examination aids or lend them to each other
- Do not communicate with any other person by any means

During written examinations, the following actions will in any case be deemed to constitute fraud or attempted fraud:

- using another person's proof of identity/campus card (student identity card)
- having a mobile telephone or any other type of media-carrying device on your desk or in your clothes
- using, or attempting to use, unauthorized resources and aids, such as the internet, a mobile telephone, smartwatch, smart glasses etc.
- having any paper at hand other than that provided by TU/e, unless stated otherwise
- copying (in any form)
- visiting the toilet (or going outside) without permission or supervision

You can start the exam now, good luck!

Formulas

Antiderivatives (without "+c")

$f(x) = F'(x)$	$F = \int f(x) dx$	$f(x) = F'(x)$	$F = \int f(x) dx$
$x^n, n \neq -1$	$\frac{x^{n+1}}{n+1}$	$\frac{\alpha x}{(1+\alpha x^2)^{\frac{1}{2}}}$	$(1+\alpha x^2)^{\frac{1}{2}}$
$\frac{1}{x}$	$\ln(x)$	$\frac{1}{(1+\alpha x^2)^{\frac{3}{2}}}$	$\frac{x}{(1+\alpha x^2)^{\frac{1}{2}}}$
e^x	e^x	$a^x, a > 0, a \neq 1$	$\frac{a^x}{\ln(a)}$
$\sin(x)$	$-\cos(x)$	$\frac{1}{\sqrt{1-x^2}}$	$\arcsin(x)$
$\cos(x)$	$\sin(x)$	$-\frac{1}{\sqrt{1-x^2}}$	$\arccos(x)$
$\frac{1}{\cos^2(x)}$	$\tan(x)$	$\frac{1}{1+x^2}$	$\arctan(x)$

Vectors and matrices

Let $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ be vectors in $\mathbb{R}^3$.

Inner / Dot product: $\mathbf{a} \bullet \mathbf{b} = \mathbf{a}^T \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$

Cross product: $\mathbf{a} \times \mathbf{b} = \det \begin{pmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{pmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix} = -(\mathbf{b} \times \mathbf{a})$.

Specifically, $\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}, \hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}, \hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}$.

Determinant of a matrix $\mathbf{A} := \{a_{ij}\}_{1 \leq i, j \leq 3}$:

$$\det(\mathbf{A}) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}).$$

Coordinate transformations

Cylindrical coordinates: For $r \geq 0, \theta \in [0, 2\pi]$: $x = r \cos(\theta), y = r \sin(\theta), z = z$, while $\det \left(\frac{\partial(x,y,z)}{\partial(r,\theta,z)} \right) = r$.

Spherical coordinates: For $R \geq 0, \theta \in [0, 2\pi], \phi \in [0, \pi]$:

$x = R \sin(\phi) \cos(\theta), y = R \sin(\phi) \sin(\theta), z = R \cos(\phi)$, while $\det \left(\frac{\partial(x,y,z)}{\partial(R,\phi,\theta)} \right) = R^2 \sin(\phi)$.

Elliptic coordinates: For describing $\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$: $x = a u \cos(v), y = b u \sin(v)$

Different integrals

Line integrals: let $\mathbf{r}(t) = x(t)\hat{\mathbf{i}} + y(t)\hat{\mathbf{j}} + z(t)\hat{\mathbf{k}}$ with $a \leq t \leq b$ be a parameter description of the curve $\mathcal{C}$.

Line integral of a scalar field f : $\int_{\mathcal{C}} f(\mathbf{r}) ds = \int_a^b f(\mathbf{r}(t)) \left| \frac{d\mathbf{r}}{dt} \right| dt$

Line integral of a vector field $\mathbf{f}$: $\int_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r} = \int_a^b \mathbf{f}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} dt$

Surface integrals: let $\mathbf{r}(u, v) = x(u, v)\hat{\mathbf{i}} + y(u, v)\hat{\mathbf{j}} + z(u, v)\hat{\mathbf{k}}$ be a parametric description of the surface $\mathcal{S}$ for $(u, v) \in \mathcal{D}$.

Surface integral of a scalar field f : $\iint_{\mathcal{S}} f(\mathbf{r}) dS = \iint_{\mathcal{D}} f(\mathbf{r}(u, v)) \left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right| du dv$

Flux of a vector field across an oriented surface $\mathbf{f}$: $\iint_{\mathcal{S}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{S} = \iint_{\mathcal{D}} \mathbf{f}(\mathbf{r}(u, v)) \cdot \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du dv$

Volume integrals:

For a variable transformation $(x, y, z) \mapsto (u, v, w)$ such that $x = x(u, v, w)$, $y = y(u, v, w)$, $z = z(u, v, w)$, let a domain $\mathcal{D}$ in xyz -space transform into a domain $\mathcal{V}$ in uvw -space. Then, the volume integral changes by the rule

$$\iiint_{\mathcal{D}} f(x, y, z) dx dy dz = \iiint_{\mathcal{V}} f(x(u, v, w), y(u, v, w), z(u, v, w)) \left| \det \left(\frac{\partial(x, y, z)}{\partial(u, v, w)} \right) \right| du dv dw$$

Gradient, Divergence, Curl, and Jacobian

Gradient: For $f : \mathcal{D} \rightarrow \mathbb{R}$ we have: $\nabla f = \frac{\partial f}{\partial x} \hat{\mathbf{i}} + \frac{\partial f}{\partial y} \hat{\mathbf{j}} + \frac{\partial f}{\partial z} \hat{\mathbf{k}}$

Divergence: For $\mathbf{f} : \mathcal{D} \rightarrow \mathbb{R}^3$ we have: $\nabla \cdot \mathbf{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$

Curl: For $\mathbf{f} : \mathcal{D} \rightarrow \mathbb{R}^3$ we have: $\nabla \times \mathbf{f} = \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) \hat{\mathbf{i}} + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) \hat{\mathbf{j}} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \hat{\mathbf{k}}$

Jacobian: For $\mathbf{f} = (f_1, f_2, f_3) : \mathcal{D} \rightarrow \mathbb{R}^3$, we have: $D\mathbf{f} = \frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)} = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial y} & \frac{\partial f_3}{\partial z} \end{pmatrix}$

Question 1

Consider the function $f(x, y) = 1 - \sqrt{xy - y^2}$.

- 4p **1a** What is the maximal domain of definition $\mathcal{D} \subseteq \mathbb{R}^2$ of the function f ? Explain your answer and sketch the domain $\mathcal{D}$.

[illegible]

- 2p **1b** Is the point $(0,0)$ an isolated point of the domain $\mathcal{D}$? Justify your answer.

[Note: Isolated points $\mathbf{x}_0 \in \mathcal{D}$ are such that for some $r > 0$, $B_r(\mathbf{x}_0) \cap \mathcal{D} = \{\mathbf{x}_0\}$. In other words, there exists a circle around $\mathbf{x}_0$ which does not contain any other point in $\mathcal{D}$ other than $\mathbf{x}_0$.]

2p **1c** What is the value of $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$?

3p **1d** Calculate the gradient of the function f , i.e. $\nabla f(x,y)$.

3p **1e** Are $\partial_x f(x, y)$ and $\partial_y f(x, y)$ continuous at $(0,0)$? Justify your answers.

3p

[illegible]

Consider the vector field $\mathbf{f}(x, y, z) = yz\hat{\mathbf{i}} + zx\hat{\mathbf{j}} + xy\hat{\mathbf{k}}$ for $x, y, z \in \mathbb{R}$.

5p

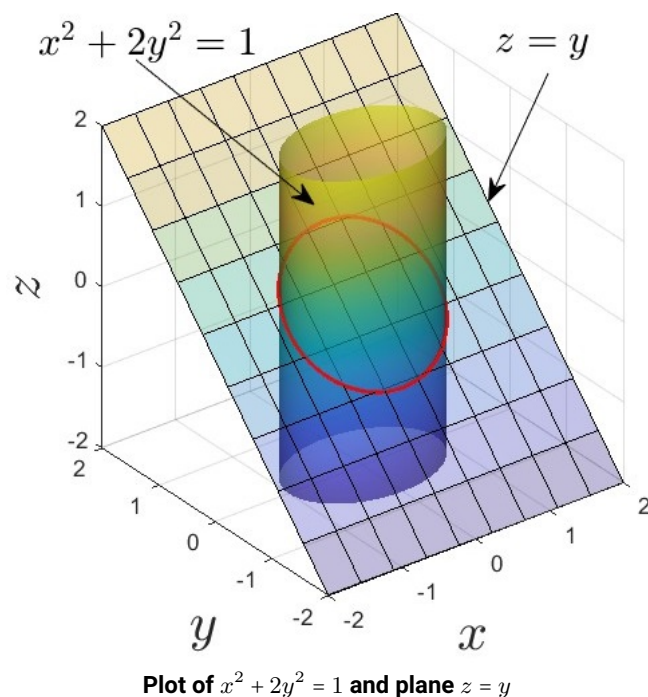
[illegible]

5p

[The flow-lines are curves determined by $\frac{dx}{f_1(x,y,z)} = \frac{dy}{f_2(x,y,z)} = \frac{dz}{f_3(x,y,z)}$]

3p **2c** What is the unit-normal to the equipotential surface of φ (i.e., level set of φ) at $(1, 2, 1)$? Explain in detail.

[Hint: you do not need φ to answer this question]



- 7p **2d** Consider the curve $\mathcal{C}$ from the point $(-1, 0, 0)$ to the point $(1, 0, 0)$ along the intersection of the elliptic cylinder $x^2 + 2y^2 = 1$ and the plane $z = y$, keeping $y \geq 0$ (see above figure).

What is the arc-length of this curve $\mathcal{C}$, and what is the value of $\int_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r}$

[Hint: You could consider using a parametrization $x = -\cos(t)$ with t in an appropriate range to describe the curve]

[illegible]

Consider the following surface (the so-called "hyperboloid of one-sheet") for $a, b > 0$:

Consider the following variable substitution $(x, y, z) \rightarrow (u, v, w)$ such that:

$$x = au\sqrt{1+w^2}\cos(v)$$

$$y = bu\sqrt{1 + w^2} \sin(v)$$

$$z = w$$

3p **3a** Determine the bounds on u, v, w describing the domain

$$\mathcal{D} := \left\{ (x, y, z) \in \mathbb{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 + z^2, \quad -1 \leq z \leq 1 \right\}$$

[illegible]

3b

$$\frac{\partial(x, y, z)}{\partial(u, v, w)}.$$
$$ab(1 + w^2)u.$$
[illegible]

5p

[Hint: Use the results in part (a) and (b).]

[illegible]

Evaluate the following integrals

8p

Determine $\iiint_D (x^2 + y^2) \, dx \, dy \, dz$.

[Hint: Use a transformation to a suitable coordinate system]

[illegible]

[illegible]

4b Flux through a surface $\iint_S \mathbf{f} \cdot d\mathbf{S}$: Find the flux of $\mathbf{f}(x, y, z) = \frac{-y\hat{\mathbf{i}} + x\hat{\mathbf{j}}}{x^2 + y^2} + e^{-x^2}\hat{\mathbf{k}}$ upward through the surface S defined as

$$\mathcal{S} := \{(x, y, z) \in \mathbb{R}^3 \mid 0 \leq y \leq 1, \quad y \leq x \leq 1, \quad z = x^2 + y^2\}$$

[illegible]

[illegible]

Question 5

Let $\mathbf{c} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$, and consider the position vector $\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$. Evaluate

2p **5a** $\nabla(\mathbf{c} \cdot \mathbf{r})$

4p **5b** $\nabla \cdot (\mathbf{c} \times \mathbf{r})$

4p

5c

$$\nabla \times (\mathbf{c} \times \mathbf{r})$$

[illegible]

6 You can write your answers in the extra space here if you exceed the allotted space

[illegible]

[illegible]